

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed client communications and design document reviews for infrastructure projects, ensuring adherence to project specifications and timelines.
- Prepared and reviewed preliminary designs and detailed drawings for civil engineering projects using **AutoCAD 2D**, contributing to project planning and execution.
- Assisted Project Managers in compiling tender documents and evaluating bids, ensuring compliance with contract requirements and project scope.
- Administered construction contracts, monitoring design implementation against project schedules and budgets to ensure successful project delivery.
- Developed Python scripts to automate data extraction and reporting for engineering workflows, improving efficiency by approximately **15%**.
- Containerized Java Spring Boot applications using **Docker**, facilitating consistent deployments across development and production environments on **AWS EC2**.
- Wrote **SQL queries** against PostgreSQL databases to validate data integrity and support ad-hoc reporting needs for project tracking.

SKILLS

CAD/BIM Software – AutoCAD 2D, AutoCAD 3D, Autodesk REVIT, ArcGIS, MapInfo

Drainage & Civil Engineering Software – MicroDrainage

Project Management Tools – Microsoft Project, Microsoft Access

Engineering Principles – Civil Engineering, Structural Engineering, Building Design, Infrastructure Design

Contracts & Procurement – PwM Forms of Contract

Programming & Scripting – Python, Java, SQL

Cloud & Containerization – AWS EC2, Docker

PROJECTS

Civil Infrastructure Design Automation (HCL Technologies)

- Developed Python scripts to automate the generation of standard civil engineering drawings and reports from design parameters, reducing manual effort by an estimated **20%**.
- Integrated with **AutoCAD** via its API to programmatically create and modify drawing elements, ensuring consistency and accuracy in design outputs.
- Authored technical documentation and user guides for the automation tools, enabling wider adoption by the engineering team at HCL.

Building Information Modelling Integration (HCL Technologies)

- Explored and prototyped the integration of **AutoCAD 3D** models with a database backend using **PostgreSQL** for enhanced data management in building projects.
- Designed data schemas to capture critical building information, facilitating easier querying and reporting for project stakeholders.
- Assisted in the preparation of detailed design specifications based on BIM model data, supporting client requirements for infrastructure projects.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)